Week 2 Internship Report

Network Administration Internship Program – IT-SIMPLERA Institute

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IP Addressing (FLSM & VLSM) and VLAN Configuration

Introduction

IP addressing is an important part of computer networking. Every device on a network needs a unique IP address to communicate. In this assignment, I learned about FLSM, VLSM, and VLANs. I also practiced subnetting and IP address planning for different network requirements.

Concepts Learned This Week

During this week, I learned:

- IP Addressing, Subnetting

- FLSM & VLSM

- Network Address, Broadcast Address

- Usable Host Range

- VLAN Basics, Access & Trunk Ports.

Fixed Length Subnet Mask (FLSM)

FLSM is a subnetting technique in which all subnets have the same subnet mask and the same number of usable host addresses. It is simple to design and manage but may waste IP addresses when different networks require different numbers of hosts.

Network ID:

The Network ID is the first IP address of a subnet. It identifies the subnet and cannot be assigned to any device.

Usable IP Addresses:

Usable IP addresses are the IP addresses that can be assigned to devices such as computers, printers, and routers within a subnet.

Broadcast Address:

The Broadcast Address is the last IP address of a subnet. It is used to send data to all devices in the same subnet and cannot be assigned to a host.

FLSM Practice Problems

Problem 1: 192.168.10.0/24, Subnet the network into 4 equal subnets.

(Have to borrow 2 bits from host to network)

4 Subnets of 192.168.10.0/24 Network

--> 192.168.10.0/26 Network ID
--> 192.168.10.1/26 First usable IP
--> 192.168.10.62/26 Last Usable IP
--> 192.168.10.63/26 Broadcast IP

--> 192.168.10.64/26 Network ID
--> 192.168.10.65/26 First usable IP
--> 192.168.10.126/26 Last Usable IP
--> 192.168.10.127/26 Broadcast IP

--> 192.168.10.128/26 Network ID
--> 192.168.10.129/26 First usable IP
--> 192.168.10.190/26 Last Usable IP
--> 192.168.10.191/26 Broadcast IP

--> 192.168.10.193/26 Network ID
--> 192.168.10.194/26 First usable IP
--> 192.168.10.254/26 Last Usable IP
--> 192.168.10.255/26 Broadcast IP

Problem 2: 192.168.20.0/24, Subnet the network into 8 equal subnets.

(Have to borrow 3 bits from host to network)

8 Subnets of 192.168.10.0/24 Network

--> 192.168.20.0/27 Network ID
--> 192.168.20.1/27 First usable IP
--> 192.168.20.30/27 Last Usable IP
--> 192.168.20.31/27 Broadcast IP

--> 192.168.20.32/27 Network ID
--> 192.168.20.33/27 First usable IP
--> 192.168.20.62/27 Last Usable IP
--> 192.168.20.63/27 Broadcast IP

--> 192.168.20.64/27 Network ID
--> 192.168.20.65/27 First usable IP
--> 192.168.20.94/27 Last Usable IP
--> 192.168.20.95/27 Broadcast IP

--> 192.168.20.96/27 Network ID
--> 192.168.20.97/27 First usable IP
--> 192.168.20.126/27 Last Usable IP
--> 192.168.20.127/27 Broadcast IP

--> 192.168.20.128/27 Network ID
--> 192.168.20.129/27 First usable IP
--> 192.168.20.158/27 Last Usable IP
--> 192.168.20.159/27 Broadcast IP

--> 192.168.20.160/27 Network ID
--> 192.168.20.161/27 First usable IP
--> 192.168.20.190/27 Last Usable IP
--> 192.168.20.191/27 Broadcast IP

--> 192.168.20.192/27 Network ID
--> 192.168.20.193/27 First usable IP
--> 192.168.20.222/27 Last Usable IP
--> 192.168.20.223/27 Broadcast IP

--> 192.168.20.224/27 Network ID
--> 192.168.20.225/27 First usable IP
--> 192.168.20.254/27 Last Usable IP
--> 192.168.20.255/27 Broadcast IP

All Subnets have the same subnet mask (255.255.255.224) means its Fixed Length Subnet Mask (FLSM).

Problem 3: 192.168.30.0/24, Subnet the network into 16 equal subnets.

(Have to borrow 4 bits from host to network)

16 Subnets of 192.168.10.0/24 Network

--> 192.168.30.0/28 1st Subnet --> 192.168.30.16/28 2nd Subnet

--> 192.168.30.32/28 3rd Subnet --> 192.168.30.48/28 4th Subnet

--> 192.168.30.64/28 5th Subnet --> 192.168.30.80/28 6th Subnet

--> 192.168.30.96/28 7th Subnet --> 192.168.30.112/28 8th Subnet

--> 192.168.30.128/28 9th Subnet --> 192.168.30.144/28 10th Subnet

--> 192.168.30.160/28 12th Subnet --> 192.168.30.176/28 12th Subnet

--> 192.168.30.192/28 13th Subnet --> 192.168.30.208/28 14th Subnet

--> 192.168.30.224/28 15th Subnet --> 192.168.30.224/28 16th Subnet

Problem 4: 172.16.0.0/24, Subnet the network into 2 equal subnets.

(Have to borrow 1 bit from host to network)

2 Subnets of 172.168.0.0/24 Network

--> 172.16.0.0/25 1st Subnet --> 172.16.0.128/25 2nd Subnet

Problem 5: 172.16.1.0/24, Subnet the network into 4 equal subnets.

(Have to borrow 2 bits from host to network)

4 Subnets of 172.168.1.0/24 Network

--> 172.16.1.0/26 1st Subnet --> 172.16.1.64/26 2nd Subnet

--> 172.16.1.128/26 3rd Subnet --> 172.16.1.192/26 4th Subnet

Problem 6: 172.16.2.0/24, Subnet the network into 8 equal subnets.

(Have to borrow 3 bits from host to network)

8 Subnets of 172.16.2.0/24 Network

--> 172.16.2.0/27 1st Subnet --> 172.16.2.32/27 2nd Subnet

--> 172.16.2.64/27 3rd Subnet --> 172.16.2.96/27 4th Subnet

--> 172.16.2.128/27 5th Subnet --> 172.16.2.160/27 6th Subnet

--> 172.16.2.192/27 7th Subnet --> 172.16.2.224/27 8th Subnet

Problem 7: 10.10.10.0/24, Subnet the network into 4 equal subnets.

(Have to borrow 2 bits from host to network)

4 Subnets of 10.10.10.0/24 Network

--> 10.10.10.0/26 1st Subnet --> 10.10.10.64/26 2nd Subnet

--> 10.10.10.128/26 3rd Subnet --> 10.10.10.192/26 4th Subnet

Problem 8: 10.20.20.0/24, Subnet the network into 8 equal subnets.

(Have to borrow 3 bits from host to network)

8 Subnets of 10.10.10.0/24 Network

--> 10.20.20.0/27 1st Subnet --> 10.20.20.32/26 2nd Subnet

--> 10.20.20.64/27 3rd Subnet --> 10.20.20.96/26 4th Subnet

--> 10.20.20.128/27 5th Subnet --> 10.20.20.160/26 6th Subnet

--> 10.20.20.192/27 7th Subnet --> 10.20.20.224/26 8th Subnet

Problem 9: 192.168.100.0/24, Subnet the network into 16 equal subnets.

(Have to borrow 4 bits from host to network)

16 Subnets of 192.168.100.0/24 Network

--> 192.168.100.0/28 1st Subnet --> 192.168.100.16/28 2nd Subnet

--> 192.168.100.32/28 3rd Subnet --> 192.168.100.48/28 4th Subnet

--> 192.168.100.64/28 5th Subnet --> 192.168.100.80/28 6th Subnet

--> 192.168.100.96/28 7th Subnet --> 192.168.100.112/28 8th Subnet

--> 192.168.100.128/28 9th Subnet --> 192.168.100.144/28 10th Subnet

--> 192.168.100.160/28 12th Subnet --> 192.168.100.176/28 12th Subnet

--> 192.168.100.192/28 13th Subnet --> 192.168.100.208/28 14th Subnet

--> 192.168.100.224/28 15th Subnet --> 192.168.100.224/28 16th Subnet

Problem 10: 192.168.200.0/24, Subnet the network into 8 equal subnets.

(Have to borrow 3 bits from host to network)

8 Subnets of 192.168.200.0/24 Network

--> 192.168.200.0/27 1st Subnet --> 192.168.200.32/27 2nd Subnet

--> 192.168.200.64/27 3rd Subnet --> 192.168.200.96/27 4th Subnet

--> 192.168.200.128/27 5th Subnet --> 192.168.200.160/27 6th Subnet

--> 192.168.200.192/27 7th Subnet --> 192.168.200.224/28 8th Subnet

Conclusion: From Problem 1 to 10, we can see that every subnet of Network has the same number of hosts and has the same subnet mask. So, that is why this subnetting technique is called a Fixed length subnet mask (FLSM).

Variable Length Subnet Mask (VLSM)

VLSM is a subnetting technique that allows different subnet sizes within the same network. It provides better utilization of IP addresses by assigning subnet sizes according to the number of hosts required in each network.

VLSM Scenarios

Scenario 1: 192.168.10.0/24, Allocate IP addresses for:

- HR Department: 60 Hosts

- IT Department: 30 Hosts

- Finance Department: 12 Hosts

- Management: 6 Hosts

For HR Department:

192.168.10.0/26	Network ID
192.168.10.1/26	First Usable IP
192.168.10.62/26	Last Usable IP
192.168.10.63/26	Broadcast

For IT Department:

192.168.10.64/27	Network ID
192.168.10.65/27	First Usable IP
192.168.10.94/27	Last Usable IP
192.168.10.95/27	Broadcast

For Finance Department:

192.168.10.96/28	Network ID
192.168.10.97/28	First Usable IP
192.168.10.110/28	Last Usable IP
192.168.10.111/28	Broadcast

For Management:

192.168.10.112/29	Network ID
192.168.10.113/29	First Usable IP
192.168.10.118/29	Last Usable IP
192.168.10.119/29	Broadcast

HR got 62 Available IPs, IT got 30 Available IPs, Finance got 14 Available IPs, and Management team got 6 IPs.

Scenario 2: 172.16.0.0/24, Allocate IP addresses for:

- Sales: 100 Hosts
- Accounts: 50 Hosts
- Administration: 25 Hosts
- Server Room: 10 Hosts

For Sales:

172.16.0.0/25	Network ID
172.16.0.1/25	First Usable IP
172.16.0.126/25	Last Usable IP
172.16.0.127/25	Broadcast

For Accounts:

172.16.0.128/26	Network ID
172.16.0.129/26	First Usable IP
172.16.0.190/26	Last Usable IP
172.16.0.191/26	Broadcast

For Administration:

172.16.0.192/27	Network ID
172.16.0.193/27	First Usable IP
172.16.0.222/27	Last Usable IP
172.16.0.223/27	Broadcast

For Servers:

172.16.0.224/28	Network ID
172.16.0.225/28	First Usable IP
172.16.0.254/28	Last Usable IP
172.16.0.255/28	Broadcast

Sales got 126 Available IPs, Accounts got 62 Available IPs, Administration got 30 Available IPs, and Servers got a total of 14 Available IPs which are the perfect VLSM subnets for the above Requirements.

Scenario 3: 192.168.20.0/24, Allocate IP addresses for:

- Engineering: 120 Hosts
- HR: 60 Hosts
- Reception: 30 Hosts
- CCTV Network: 10 Hosts

For Engineering:

192.168.20.0/25 (Available Hosts: 126)

For HR:

192.168.20.128/26 (Available Hosts: 62)

For Reception:

192.168.20.192/27 (Available Hosts: 30)

For CCTV Network:

192.168.20.224/28 (Available Hosts: 16)

Scenario 4: 10.10.10.0/24, Allocate IP addresses for:

- Office A: 70 Hosts
- Office B: 35 Hosts
- Office C: 15 Hosts
- Office D: 5 Hosts

For Office A: 10.10.10.0/25 (126 Available Hosts)

For Office B: 10.10.10.128/26 (62 Available Hosts)

For Office C: 10.10.10.192/27 (30 Available Hosts)

For Office D: 10.10.10.224/29 (6 Available Hosts)

VLAN (Virtual Local Area Network)

A Virtual Local Area Network (VLAN) is a logical grouping of devices within a network. It allows devices to communicate as if they are on the same network, even if they are connected to different switches. VLANs improve network security, reduce broadcast traffic, and make network management easier.

VLANs are widely used in enterprise networks to improve security, reduce broadcast traffic, and simplify network management. Different departments, such as HR, Sales, and IT, can be placed in separate VLANs even if they are connected to the same switch.

LAB Overview:

After understanding the concepts of FLSM, VLSM, and VLANs, the next step is to implement them in a practical environment. I will use Cisco Packet Tracer and GNS3 to design and configure a network topology.

Packet Tracer Implementation

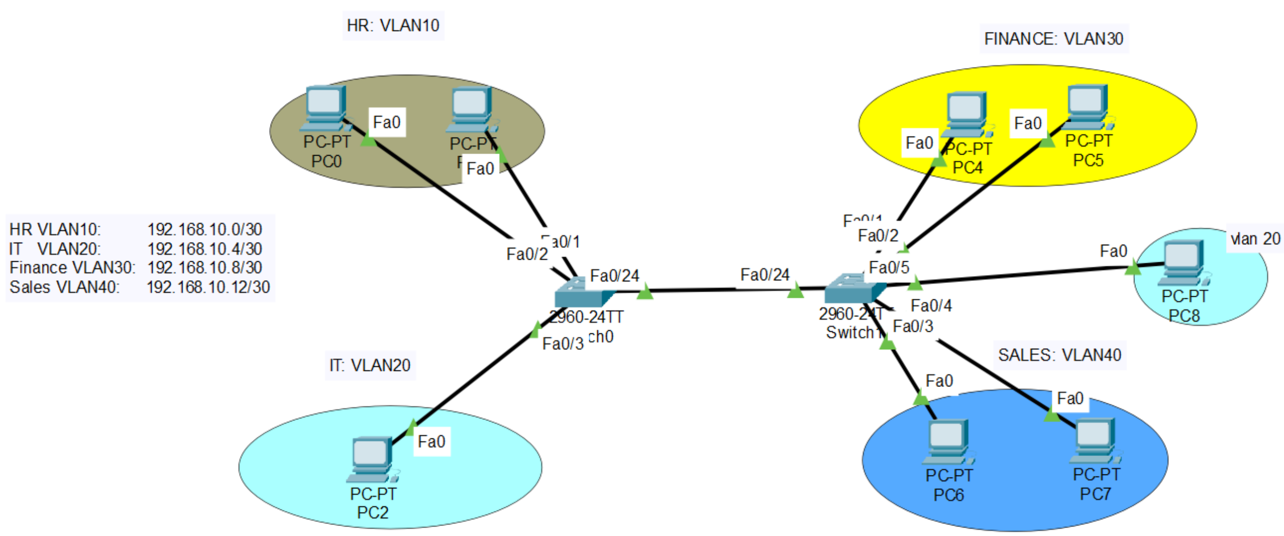
A network topology was created in Cisco Packet Tracer to demonstrate the implementation of VLANs and VLSM. The topology consists of two Cisco 2960 switches connected through a trunk link. Four VLANs were created to represent different departments within an organization.

The configured VLANs and VLSM addressing scheme used in this topology is:

VLAN	Department	Network
VLAN 10	HR	192.168.10.0/30
VLAN 20	IT	192.168.10.4/30
VLAN 30	Finance	192.168.10.8/30
VLAN 40	Sales	192.168.10.12/30

Configuration Steps:

1. Created VLANs 10, 20, 30, and 40.
2. Assigned access ports to the appropriate VLANs.
3. Configured the trunk link between the two switches.
4. Assigned IP addresses according to the VLSM plan.
5. Verified VLAN and trunk configurations.
6. Performed ping tests to verify connectivity.



```
Switch#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Gig0/1, Gig0/2
20   VLAN0020                         active    Fa0/5
30   FINANCE                          active    Fa0/1, Fa0/2
40   SALES                            active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
Switch#
Switch#
Switch#
Switch#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/24      1,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      1,20,30,40
```

PC8

Physical Config Desktop Programming Attributes

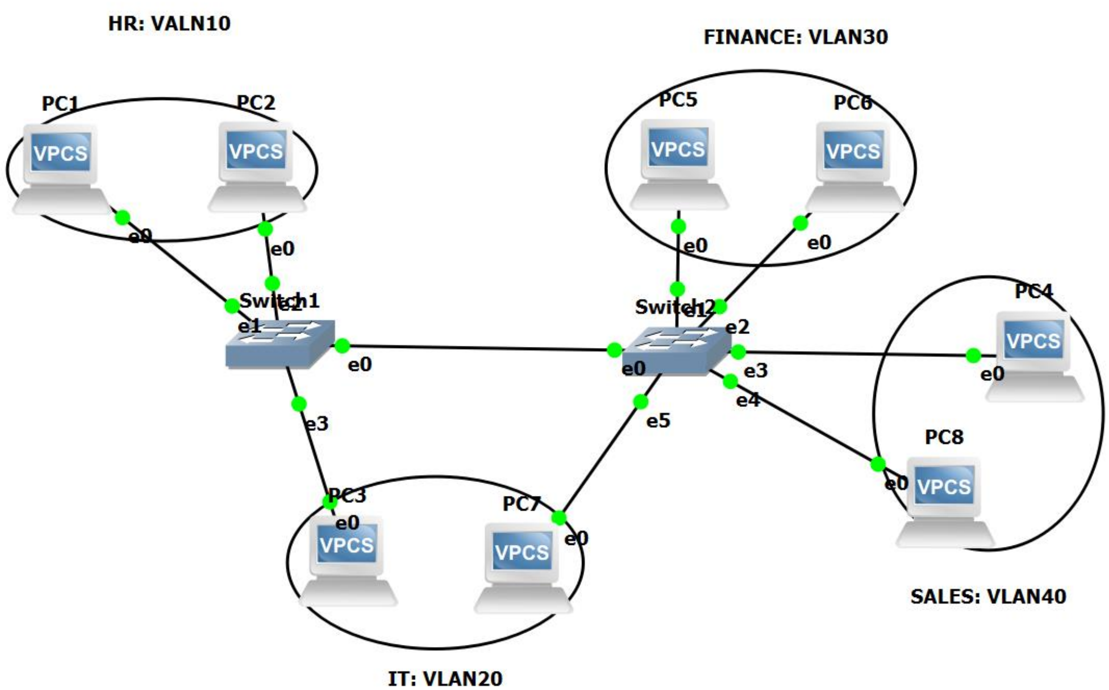
Command Prompt X

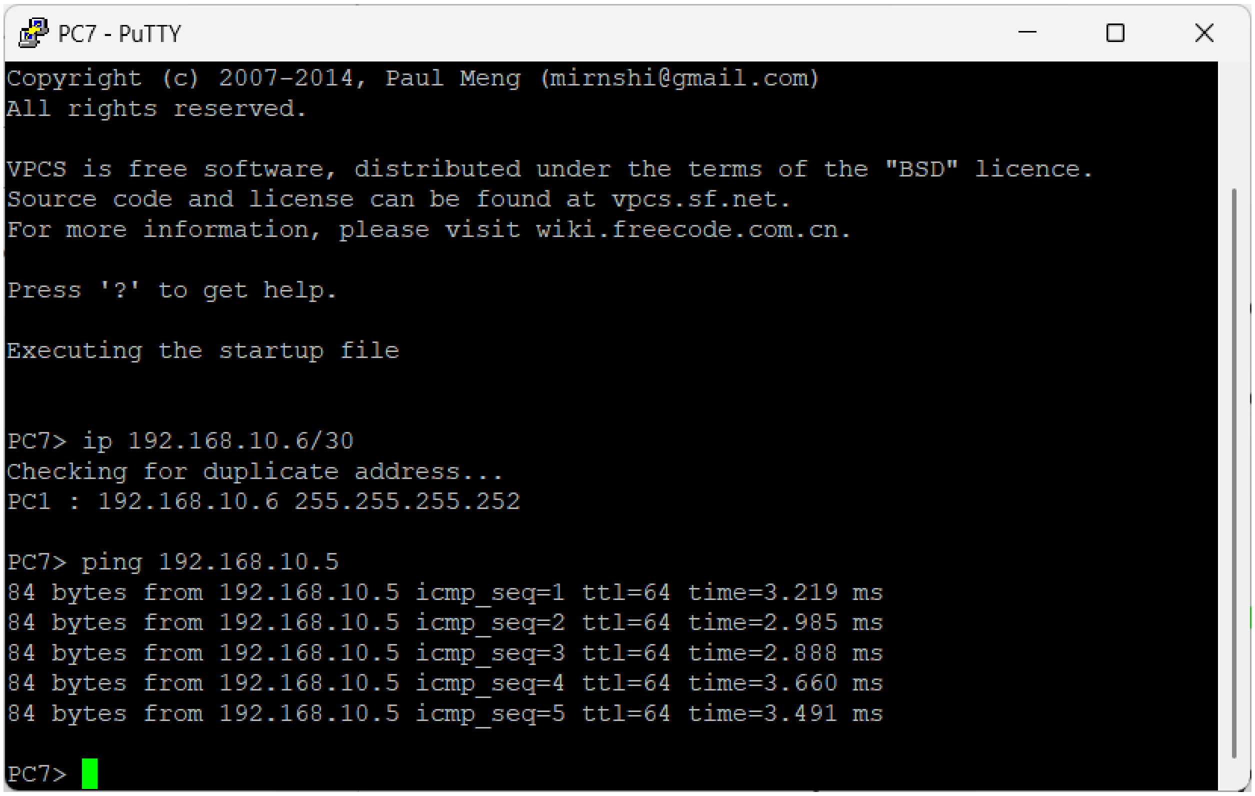
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time=3ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

GNS3 Implementation

- Recreated the same network topology in GNS3 for practical implementation.
- Added two Layer 2 switches and connected them using a trunk link.
- Created four VLANs (VLAN 10, VLAN 20, VLAN 30, and VLAN 40) and assigned the connected PCs to their respective VLANs.
- Configured access ports and assigned IP addresses according to the VLSM addressing plan.
- Verified the network configuration using show vlan brief, show interfaces trunk, and ping tests to ensure proper VLAN functionality and connectivity.

VLAN	Department	Network
VLAN 10	HR	192.168.10.0/30
VLAN 20	IT	192.168.10.4/30
VLAN 30	Finance	192.168.10.8/30
VLAN 40	Sales	192.168.10.12/30





Conclusion

This project provided practical experience in IP addressing, subnetting, and VLAN configuration. FLSM and VLSM were used to design an efficient IP addressing scheme, while VLANs were configured to segment the network. The configuration was successfully verified using Cisco IOS commands and connectivity tests. Overall, this assignment improved my understanding of network design and basic switch configuration.

Learning Outcomes

Understood the difference between FLSM and VLSM.

Performed subnet calculations and IP allocation.

Configured VLANs and assigned switch ports.

Configured and verified trunk links.

Verified the network using Cisco IOS commands and ping tests.

Improved practical skills in Cisco Packet Tracer and GNS3.